

Continuum-Based Limits of KPI-Driven Analysis in Enterprise Systems

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Abstract

We address a single scope-locked question: what are the formal limits of KPI-based enterprise analysis when enterprises are modeled as continua K_{EA} under the Ontology of Continua (OC) and its executable theory specification `ETS.K_EA.v1.1`. Within this framework, KPIs are treated as admissible readout maps defined on the OC/ETS observable domain $\Omega_{\text{obs}} \subseteq \Omega(K_{EA})$, where observability itself is a structural property of the enterprise continuum, comprising an axis-level observability component, a factor S_{obs} , and an ObservabilityInvariant.

KPI-based observation is formalized as a quotienting of Ω_{obs} by the fibers of the KPI readout map. On this basis, we establish existence and genericity results showing that KPI observation is intrinsically non-identifying: structurally distinct enterprise states may remain indistinguishable under any finite admissible KPI family, regardless of sampling resolution, aggregation schemes, or downstream deterministic statistical post-processing.

These results are strictly diagnostic. They delineate fundamental information-theoretic limits of KPI-driven analysis and characterize what cannot be recovered even in principle in continuum-modeled enterprise systems.

Keywords: enterprise continuum; KPI observability; identifiability limits; non-identifiability; impossibility results

1 Problem Statement

This preprint addresses exactly one scope-locked question:

Q5: What are the formal limits of KPI-based enterprise analysis in continuum-modeled systems K_{EA} ?

Hard constraints. All statements are derived strictly within the Ontology of Continua (OC) and the executable theory specification `ETS.K_EA.v1.1`. No new primitives are introduced, and no reinterpretation or extension of OC or ETS is permitted. The scope is purely diagnostic: the results are negative in nature and establish existence and impossibility claims concerning what KPI-based analysis cannot identify.

Explicit non-claims. This work makes no managerial, governance, optimization, or control claims. It does not recommend KPIs, prescribe dashboards, propose decision rules, or advance performance narratives. No probabilistic or statistical enrichment is introduced beyond what is already contained in OC/ETS. References to aggregation, sampling, or post-processing are included solely to emphasize that the derived limits are structural and do not disappear under such operations.

Core distinction. The central distinction formalized in this work is between *observability* and *identifiability*. Observability concerns whether readouts are defined and admissible on a given domain, whereas identifiability concerns whether those readouts uniquely determine the underlying system state. This distinction is classical in systems theory and system identification [2, 3, 6]. While OC/ETS explicitly provides observability structures at the level of the enterprise continuum, the question addressed here is whether KPI-based readouts can be identifying in the formal sense. The main result is that, in continuum-modeled enterprise systems, KPI-based observation is generically non-identifying.

Formal relevance. Under OC/ETS, enterprises are modeled as continua with state space $\Omega(K_{EA})$, structured by axes, flows, thresholds, cycles, and a continuum measure. KPI readouts are, by construction, finite-dimensional projections of this state space. It is a standard result in identification theory that such projections generally induce equivalence classes of indistinguishable states unless injectivity is explicitly enforced [3, 5]. This preprint formalizes that reduction in the OC/ETS setting and shows that it induces unavoidable non-identifiability: multiple structurally distinct enterprise states can share identical KPI values.

2 Formal Setting

Throughout this preprint we assume the Ontology of Continua (OC) together with the frozen executable theory specification `ETS.K_EA.v1.1`. An enterprise is modeled as a continuum K_{EA} with state space $\Omega(K_{EA})$, boundary $\partial\Omega(K_{EA})$, axes A , flows J , thresholds Θ , cycles C , and a continuum measure k_{EA} . No primitives beyond those defined in OC and ETS are introduced or assumed.

2.1 ETS-anchored observability structures

In `ETS.K_EA.v1.1`, observability is not treated as an external modeling assumption but as an internal structural component of the enterprise continuum. In particular, the executable specification includes:

- an information-axis component *observability*, located within the information/semantics axis block;
- a factor S_{obs} in the factorization of the continuum measure k_{EA} ;
- an explicit *ObservabilityInvariant* listed among the system invariants;
- and a phase-logic condition `observability_lost` acting as a STOP trigger.

Observation is therefore defined only on the OC/ETS-admissible observable domain

$$\Omega_{\text{obs}} \subseteq \Omega(K_{EA}),$$

consisting of all enterprise states for which S_{obs} is defined and the *ObservabilityInvariant* holds. When the ETS phase condition `observability_lost` applies, KPI readout is outside scope and no observation map is assumed to be defined.

2.2 Observation as structural quotienting

Fix a finite KPI family $K = (\kappa_1, \dots, \kappa_m)$ admissible under OC/ETS. Each KPI κ_i is treated as a readout map $\kappa_i : \Omega_{\text{obs}} \rightarrow \mathbb{R}$. Together they define the KPI readout map

$$\pi_K : \Omega_{\text{obs}} \rightarrow \Sigma_{\text{obs}} \subseteq \mathbb{R}^m, \quad \pi_K(\omega) = (\kappa_1(\omega), \dots, \kappa_m(\omega)).$$

As in classical observation theory, the map π_K induces an equivalence relation $\sim_K$ on Ω_{obs} defined by

$$\omega_1 \sim_K \omega_2 \iff \pi_K(\omega_1) = \pi_K(\omega_2).$$

Accordingly, KPI-based observation partitions Ω_{obs} into fibers (preimages) of π_K . The observation space Σ_{obs} is therefore identified with the quotient

$$\Sigma_{\text{obs}} \cong \Omega_{\text{obs}} / \sim_K,$$

which represents the maximal information preserved by the KPI readout [2, 3].

All subsequent limits established in this preprint are statements about this quotienting. Identifiability of enterprise states requires injectivity of π_K on a domain of interest. In contrast, KPI-based observation generically collapses structurally distinct states in Ω_{obs} into the same equivalence class.

3 Observation and KPI Spaces

Definition 1 (Enterprise State Space). Let $\Omega := \Omega(K_{EA})$ denote the OC/ETS enterprise state space of the enterprise continuum K_{EA} . All constructions in this section are carried out strictly within the Ontology of Continua and the frozen executable specification `ETS.K_EA.v1.1`. No additional primitives, regularity assumptions, or modeling hypotheses are introduced.

Definition 2 (Observable Domain Ω_{obs}). The executable specification `ETS.K_EA.v1.1` includes (i) an information-axis component *observability*, (ii) a factor S_{obs} in the factorization of the continuum measure k_{EA} , and (iii) an explicit *ObservabilityInvariant*. Accordingly, define the observable domain

$$\Omega_{\text{obs}} := \{\omega \in \Omega \mid S_{\text{obs}}(\omega) \text{ is defined and the ObservabilityInvariant holds}\}.$$

Only states in Ω_{obs} are admissible inputs to observation. States for which the ETS phase condition `observability_lost` applies lie outside the scope of this definition.

Definition 3 (Observation Space Σ_{obs}). Define the observation space Σ_{obs} as the codomain of all OC/ETS-admissible readout maps defined on Ω_{obs} . Once a finite KPI family $K = (\kappa_1, \dots, \kappa_m)$ is fixed, Σ_{obs} is identified with a subset of $\mathbb{R}^m$.

Definition 4 (KPI Family and Readout Map). Fix a finite KPI family $K := (\kappa_1, \dots, \kappa_m)$ such that each $\kappa_i : \Omega_{\text{obs}} \rightarrow \mathbb{R}$ is an OC/ETS-admissible readout, i.e. it is defined on Ω_{obs} and does not violate the *ObservabilityInvariant*. The induced KPI readout map is

$$\pi_K : \Omega_{\text{obs}} \rightarrow \Sigma_{\text{obs}} \subseteq \mathbb{R}^m, \quad \pi_K(\omega) = (\kappa_1(\omega), \dots, \kappa_m(\omega)).$$

No assumption of injectivity, completeness, faithfulness, sufficiency, or statistical optimality is imposed.

Definition 5 (K -Indistinguishability and Fibers). Two states $\omega_1, \omega_2 \in \Omega_{\text{obs}}$ are K -indistinguishable if and only if

$$\pi_K(\omega_1) = \pi_K(\omega_2).$$

Equivalently, they lie in the same fiber of π_K , defined as

$$\text{Fib}_K(y) := \pi_K^{-1}(y) \subseteq \Omega_{\text{obs}}, \quad y \in \Sigma_{\text{obs}}.$$

This fiber representation is standard in observation and identification theory [2, 3].

Definition 6 (Identifiability on a Domain). Let $D \subseteq \Omega_{\text{obs}}$ be any domain of interest. KPI-based observation is said to be *identifying on D* if the restricted map $\pi_K|_D$ is injective, i.e.,

$$\forall \omega_1, \omega_2 \in D : \quad \pi_K(\omega_1) = \pi_K(\omega_2) \implies \omega_1 = \omega_2.$$

If $\pi_K|_D$ fails to be injective, then there exist distinct states in D that are KPI-indistinguishable. This definition corresponds to structural identifiability in the classical system identification sense [3, 6].

Definition 7 (Quotient Induced by KPI Observation). The equivalence relation $\sim_K$ induced by KPI equality defines the quotient space

$$\Omega_{\text{obs}} / \sim_K,$$

whose elements are the KPI fibers. KPI-based observation factors through this quotient: all structural variation within a fiber is necessarily lost once observation is restricted to KPI read-outs. This loss of information is intrinsic to non-injective observation maps [1].

4 Impossibility and Non-Identifiability Results

This section establishes formal limits of KPI-based observation for the enterprise continuum K_{EA} under the Ontology of Continua and the frozen executable specification `ETS.K_EA.v1.1`. All results are strictly diagnostic and negative in nature: they delimit what KPI-based observation cannot identify even in principle.

The formal structure follows the classical observation–identification paradigm: observation is modeled as a map from a state space to an observation space, and identifiability corresponds to injectivity of that map on a domain of interest (see, e.g., classical treatments in control and system identification [2, 3, 6]). The specific contribution of the OC/ETS framework is to instantiate this paradigm for enterprise continua with explicit observability constraints.

4.1 Existence of KPI-Indistinguishable States

Theorem 1 (Non-Identifiability in the Absence of Injectivity). Let $K = (\kappa_1, \dots, \kappa_m)$ be any finite admissible KPI family with induced readout map

$$\pi_K : \Omega_{\text{obs}} \rightarrow \Sigma_{\text{obs}}.$$

If π_K is not injective on Ω_{obs} , then there exist distinct states $\omega_1 \neq \omega_2 \in \Omega_{\text{obs}}$ such that

$$\pi_K(\omega_1) = \pi_K(\omega_2).$$

Proof. Non-injectivity of π_K means, by definition, that there exist at least two distinct elements of Ω_{obs} that are mapped to the same observation. These states are therefore indistinguishable under KPI observation. $\square$

Although elementary, this statement is foundational: it makes explicit that non-identifiability is the default situation whenever injectivity of the observation map is not guaranteed.

4.2 Absence of Universal KPI Completeness

Theorem 2 (No Universal KPI Completeness). There exists no finite admissible KPI family K such that the associated readout map π_K uniquely identifies all states in Ω_{obs} , unless injectivity is imposed as an additional constraint on the admissible state space. In particular, neither OC nor `ETS.K_EA.v1.1` implies injectivity of any finite KPI readout map on Ω_{obs} .

OC/ETS-grounded argument. The Ontology of Continua and ETS.K_EA.v1.1 fix the structural components of the enterprise continuum, including its state space, admissible observable domain, axes, flows, thresholds, cycles, and the factorization of the continuum measure k_{EA} . They do not contain any axiom or admissibility condition requiring a user-chosen KPI readout map π_K to be injective on Ω_{obs} .

This mirrors the classical situation in system identification, where identifiability is not automatic but must be established or enforced through additional structural assumptions, excitation conditions, or model restrictions (see, e.g., [3, 6]). Within OC/ETS, no such enforcing assumption is present. Hence universal KPI completeness cannot be concluded from OC/ETS alone. $\square$

4.3 Structural Indistinguishability as an Intrinsic Feature

Proposition 1 (Loss of Structural Information Along KPI Fibers). Let K be any admissible KPI family and let $\omega_1, \omega_2 \in \Omega_{\text{obs}}$ lie in the same fiber of π_K . Then all OC/ETS-structural information distinguishing ω_1 from ω_2 is unrecoverable from KPI values alone.

Proof. If ω_1 and ω_2 lie in the same fiber, then $\pi_K(\omega_1) = \pi_K(\omega_2)$. Any downstream procedure whose input depends only on KPI values receives identical input for ω_1 and ω_2 . Consequently, all structural variation within a fiber is eliminated once observation factors through the quotient $\Omega_{\text{obs}}/\sim_K$. $\square$

This fiber-based loss of information is the direct analogue of classical projection-induced ambiguity in partially observed systems (cf. indistinguishable states in linear and nonlinear observation theory [2, 4]).

4.4 Independence from Statistical or Algorithmic Enrichment

Proposition 2 (Post-Processing Cannot Restore Identifiability). Let $g : \Sigma_{\text{obs}} \rightarrow Z$ be any deterministic post-processing map, including aggregation, smoothing, scoring, embedding, or transformation. Then the composite map $g \circ \pi_K$ is identifying on a domain $D \subseteq \Omega_{\text{obs}}$ only if π_K is identifying on D .

Proof. If $\pi_K(\omega_1) = \pi_K(\omega_2)$, then $(g \circ \pi_K)(\omega_1) = (g \circ \pi_K)(\omega_2)$ for any deterministic function g . Thus post-processing cannot separate states that are already KPI-indistinguishable. Identifiability of $g \circ \pi_K$ therefore implies identifiability of π_K on the same domain. $\square$

This reflects a standard principle in identification theory: no deterministic transformation of insufficient data can recover information that was never observed in the first place [3, 5].

4.5 Observability Versus Identifiability

Proposition 3 (Observability Does Not Imply Identifiability). ETS-level observability guarantees only that KPI readouts are defined on Ω_{obs} . It does not imply that KPI readouts uniquely identify states in Ω_{obs} .

Proof. Observability, as used here, is an admissibility condition ensuring that readouts are defined on the domain Ω_{obs} . Identifiability is a distinct property, equivalent to injectivity of the readout map on a domain of interest. OC and ETS.K_EA.v1.1 do not equate these notions and do not enforce injectivity. This separation mirrors the classical distinction between observability and identifiability in control and system identification theory [2, 3]. $\square$

5 Proof Sketches and Technical Remarks

This section provides concise technical remarks clarifying the logical status and scope of the results established in Section 4. The purpose is not to introduce additional assumptions, strengthen theorems, or extend proofs, but to make explicit what follows strictly from the Ontology of Continua (OC) and the frozen executable specification `ETS.K.EA.v1.1`, and what does not.

5.1 Source of the Impossibility Results

The fundamental structural fact underlying all impossibility statements is that KPI-based observation is represented by a map

$$\pi_K : \Omega_{\text{obs}} \rightarrow \Sigma_{\text{obs}},$$

which necessarily factors through the quotient $\Omega_{\text{obs}}/\sim_K$.

Once observation is expressed in this form, non-identifiability follows immediately: any two distinct states lying in the same fiber of π_K are indistinguishable by construction. This argument is purely set-theoretic and logical. No assumptions about smoothness, topology, dimensionality, or regularity of Ω_{obs} are required.

In this sense, the impossibility results here are structurally analogous to classical observation-induced ambiguities in control and identification theory, where indistinguishability arises from projection onto a lower-information output space (cf. [2, 3]).

5.2 On Injectivity and Admissible Assumptions

Many classical impossibility or non-identifiability results in system identification are derived under auxiliary assumptions on the observation map, such as genericity, excitation conditions, smoothness, or dimensional mismatch between state and output spaces [3, 6].

In contrast, this preprint deliberately avoids importing any such assumptions as new primitives. Doing so would exceed the OC/ETS scope and undermine the diagnostic character of the results.

Accordingly, all statements are formulated in the minimal logically correct form:

- if the KPI readout map π_K is not injective, then distinct KPI-indistinguishable states exist (Theorem 1);
- OC and `ETS.K.EA.v1.1` do not imply injectivity of any admissible finite KPI family, and therefore do not support claims of universal KPI completeness (Theorem 2);
- any deterministic transformation applied to KPI readouts preserves non-identifiability (Proposition 2).

This framing reflects the intended scope: a formal diagnostic boundary, not an engineering guideline or modeling prescription.

5.3 Observable Domain and Loss of Observability

The observable domain Ω_{obs} is a strict subset of the full enterprise state space $\Omega(K_{EA})$. It is defined exclusively via ETS-level observability structures, namely the existence of the factor S_{obs} and the validity of the *ObservabilityInvariant*.

States for which the ETS phase condition `observability_lost` applies are explicitly outside the scope of this analysis. In such states, KPI readouts are not assumed to be defined, and no claims about identifiability or non-identifiability are made. All impossibility results in this preprint are therefore conditional on admissible observability.

This separation mirrors standard distinctions in control theory, where questions of identifiability are only meaningful on domains where outputs are well-defined [2].

5.4 Why Sampling and Aggregation Cannot Help

Any aggregation over time windows, smoothing, scoring, embedding, or similar analytic operation applied to KPI values can be represented as a deterministic map

$$g : \Sigma_{\text{obs}} \rightarrow Z$$

acting on the KPI readout $\pi_K(\omega)$.

Deterministic post-processing cannot refine the partition induced by π_K : if two states yield identical KPI vectors, then all deterministic functions of those vectors are also identical. Consequently, no increase in sampling frequency, no aggregation strategy, and no deterministic statistical enrichment can restore identifiability once it is lost at the level of KPI readout.

This is a standard result in identification theory: information that is not present in the observation cannot be reconstructed by downstream processing alone [3, 5].

5.5 Interpretation of the Results

The results established in this preprint are not claims about the usefulness, quality, or desirability of KPIs. They are strictly formal statements about limits of identifiability.

Within OC and ETS.K_EA.v1.1, KPI-based observation modeled as a finite readout map cannot be assumed to uniquely identify the underlying enterprise state without introducing additional identifying structure or constraints that lie outside the OC/ETS framework.

Read in this way, the results function as a clarification of scope: they specify what KPI-based analysis can and cannot be expected to recover at the level of formal enterprise continua.

6 Scope, Assumptions, and Falsifiability

6.1 Scope of the Preprint

This preprint has a deliberately narrow and explicitly delimited scope. It is a diagnostic analysis of *KPI-based observation* for enterprise continua modeled under the Ontology of Continua (OC) and the frozen executable theory specification ETS.K_EA.v1.1.

We do not propose a new enterprise model, a new KPI methodology, nor a new estimation, optimization, governance, or decision framework. Instead, we make explicit structural consequences that already follow from the OC/ETS setup itself:

- the enterprise is represented as an OC/ETS continuum with state space $\Omega(K_{EA})$ and an admissible observable domain $\Omega_{\text{obs}} \subseteq \Omega(K_{EA})$;
- KPI observation is modeled as a finite readout map $\pi_K : \Omega_{\text{obs}} \rightarrow \Sigma_{\text{obs}} \subseteq \mathbb{R}^m$;
- identifiability is defined strictly as injectivity of π_K on a specified domain of interest.

The central conclusion is not that KPIs are ineffective, misleading, or inappropriate, but that identifiability cannot be inferred from OC/ETS alone, and that deterministic post-processing cannot recover information already lost at the level of KPI readout.

6.2 Explicit Assumptions

All results in this preprint rely exclusively on the following explicit and exhaustive assumptions:

1. **OC/ETS state space.** The enterprise continuum K_{EA} is described by OC primitives (axes, flows, thresholds, cycles, and a continuum measure), with state space $\Omega(K_{EA})$. Observation is restricted to the admissible domain Ω_{obs} , defined solely by ETS-level observability structures.

2. **Finite KPI observation.** A KPI family $K = (\kappa_1, \dots, \kappa_m)$ is finite and admissible under `ETS.K_EA.v1.1`, inducing a readout map $\pi_K : \Omega_{\text{obs}} \rightarrow \Sigma_{\text{obs}} \subseteq \mathbb{R}^m$.
3. **No hidden enrichment.** Any analytic, statistical, or computational procedure that operates only on KPI values is modeled as a deterministic post-processing map composed with π_K .

No assumptions of smoothness, continuity, dimensionality, genericity, regularity, noise structure, or probabilistic modeling are required for the logical core of the results in Section 4.

6.3 What Is Explicitly Not Claimed

This preprint explicitly does *not* claim:

- that KPI observation cannot be identifying on restricted domains $D \subseteq \Omega_{\text{obs}}$;
- that identifiability cannot be achieved by introducing additional observables, sensors, experimental interventions, or structural constraints beyond OC/ETS;
- that any concrete KPI system used in practice is necessarily non-identifying or analytically deficient.

The claim is strictly limited to the following statement: OC and `ETS.K_EA.v1.1` alone do not entail injectivity of an arbitrary finite KPI readout map, and deterministic post-processing cannot recover information lost within KPI fibers.

6.4 Falsifiability and Empirical Interface

Because the results of this preprint are diagnostic and largely negative, falsifiability must be interpreted at the appropriate formal and empirical interfaces.

1. **Formal falsification (model level).** If OC or `ETS.K_EA.v1.1` were extended by an explicit axiom enforcing injectivity of π_K on Ω_{obs} for a specified class of KPI families, then Theorem 2 would no longer hold within that amended formal system. This would constitute a formal falsification of the scope of the present results.
2. **Empirical falsification (measurement level).** For a fixed enterprise, a fixed KPI family K , and a fixed readout protocol, non-identifiability can be empirically witnessed by exhibiting two distinct operationally realizable states $\omega_1 \neq \omega_2 \in \Omega_{\text{obs}}$ such that $\pi_K(\omega_1) = \pi_K(\omega_2)$. Existence of such a pair directly demonstrates non-identifiability of K on the tested domain. Conversely, failure to find such pairs does not establish global injectivity; it only supports identifiability for the explored domain and protocol.
3. **Protocol dependence.** Because KPI values depend on sampling, aggregation windows, and cadence, any empirical claim must fix the readout protocol explicitly. Once fixed, the protocol is absorbed into the definition of π_K , and the formal results apply unchanged.

Accordingly, the falsifiable content of this preprint can be summarized as follows: *for a fixed KPI readout protocol, if two distinct admissible enterprise states yield identical KPI values, then KPI-based observation is non-identifying on that tested domain.* This statement is a direct consequence of the definitions in Section 2.

7 Literature Positioning

The formal structure employed in this preprint follows a well-established and widely used paradigm in the mathematical treatment of observation and identification. A system is characterized by a state space, observation is represented as a map from states to an observation space, and identifiability corresponds to injectivity of that map on a domain of interest. This abstract structure underlies classical work in control theory and system identification and serves as the baseline against which the present results are positioned.

7.1 Observation as a Map and Loss of Information

Modeling observation as a map $\pi : \Omega \rightarrow \Sigma$ and analyzing the induced equivalence relation on the state space is classical. Observation partitions Ω into fibers (preimages) of π , and any two states lying in the same fiber are observationally indistinguishable under the given observation scheme. This perspective is already implicit in early state-space formulations of control systems and observability [2].

In control-theoretic and systems-theoretic language, this corresponds to the standard distinction between *observability* and *identifiability*. Observability concerns whether outputs are defined and informative in principle, whereas identifiability is the stronger property that states or parameters are uniquely determined by outputs under specified conditions. The fact that observability alone does not guarantee identifiability is well-known and widely recognized in the literature.

7.2 Identifiability in System Identification

The system identification literature traditionally distinguishes between two notions of identifiability. *Structural identifiability* refers to uniqueness under idealized, noise-free observations within a given model class, while *practical identifiability* concerns uniqueness under finite data, noise, and experimental limitations. This distinction is classical and extensively documented [3, 6].

The results of this preprint operate entirely at the structural level. They formalize the fact that, within the Ontology of Continua and the frozen executable specification `ETS.K_EA.v1.1`, no general structural identifiability guarantee follows for arbitrary finite KPI readout maps. This is a statement about what is *not implied* by the modeling framework itself, independent of estimation algorithms, noise models, or data volume. In this respect, the results are fully aligned with standard identifiability theory rather than in tension with it.

7.3 OC/ETS-Specific Contribution

The only nonstandard element in the present analysis is the enterprise modeling substrate. OC and `ETS.K_EA.v1.1` provide a formally specified enterprise continuum with a state space $\Omega(K_{EA})$, explicit structural components (axes, flows, thresholds, cycles), and a factorized continuum measure. In addition, ETS constrains admissible observation through explicit observability conditions and invariants.

Within this substrate, KPI observation becomes a concrete instantiation of the general observation-map paradigm:

$$\pi_K : \Omega_{\text{obs}} \rightarrow \Sigma_{\text{obs}} \subseteq \mathbb{R}^m.$$

Once this structure is fixed, the non-identifiability results follow directly, without importing auxiliary assumptions such as smoothness, dimensionality arguments, or genericity conditions. The contribution of this preprint is therefore not a new impossibility theorem, but the explicit localization of standard identifiability limits within the OC/ETS enterprise continuum framework.

7.4 Relation to Partial Observability and Representation Limits

In many applied domains, including control, econometrics, and organizational measurement, only partial projections of the underlying system are observable. KPI dashboards are an archetypal example: they compress complex enterprise states into a finite collection of scalar outputs.

From this perspective, indistinguishability of distinct underlying states is not an anomaly but an expected consequence of projection. Identifiability typically requires additional structure, such as richer sensing, persistent excitation, experimental intervention, restrictive model classes, or explicit injectivity constraints [4, 5].

OC/ETS makes this separation explicit by distinguishing:

- admissibility of observation via the observable domain Ω_{obs} ,
- the readout mechanism π_K ,
- and the uniqueness question captured by injectivity and identifiability.

The present note should therefore be read as a formal reminder that *projection-induced ambiguity is intrinsic* and does not disappear through aggregation, smoothing, or deterministic post-processing alone.

7.5 Positioning of the Present Results

We draw on classical literature in observability, system identification, and identifiability to anchor the formal notions used here [2, 3, 6]. No attempt is made to provide a comprehensive survey. The purpose of this section is positioning rather than comparison: the logical structure of the results is standard, while their application to KPI-based enterprise observation within the OC/ETS continuum framework is specific and diagnostic in nature.

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